JESSE WOOD

Postdoctoral Research Fellow

AI for analytical chemistry

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SUMMARY

Postdoctoral Research Fellow at Te Herenga Waka—Victoria University of Wellington, working where machine learning meets analytical chemistry and Indigenous knowledge. PhD in Artificial Intelligence (2026) on deep learning for mass spectrometry in seafood authentication, with Transformer, mixture-of-experts and self-supervised contrastive methods that set new benchmarks over traditional chemometrics. BE (Hons) in Software Engineering and four years building production data services at NIWA. I write research software that ships: open-source tools for literature management, provenance and GPU scheduling that my group uses every day.

EXPERIENCE

Postdoctoral Research Fellow - *Victoria University of Wellington* 2026 – Present

- **Hiringa te Mahara:** researcher in the Aotearoa New Zealand pod (hiringatemahara.nz) of Abundant Intelligences (abundant-intelligences.net), an Indigenous-led international programme reimagining AI from Indigenous knowledge systems. Based in the School of Engineering and Computer Science, supervised by Dr Kevin Shedlock.
- **Species authentication:** deep learning on GC-MS chromatograms to verify fish species, framed through kaitiakitanga and aimed at protecting marine resources from illegal, unreported and unregulated (IUU) fishing. Ongoing collaboration with the Bioeconomy Science Institute (formerly Plant & Food Research).
- **Research software:** multi-batch retention-time alignment (*joint-align*), self-supervised pretraining on MoNA and MassBank reference spectra (*chroma-dcnn*), and open-set detection with conformal guarantees for unseen species (*red-herring*). All with CI, typed Python and enforced test coverage.
- **Teaching:** lecturing and curriculum development for undergraduate Python programming and AI/ML courses.
- **Service:** elected Postdoctoral Researchers Representative on the board of the AI Research Association of Aotearoa New Zealand (AIRA).

Research Assistant - *Victoria University of Wellington* 2025 – 2026

- Continued the doctoral research on AI for marine biomass analysis with the VUW machine learning groups and industry partners.

Software Developer Intern, then Contract Software Engineer - *NIWA* 2020 – 2024

- Built and deployed end-to-end web services (Linux, PostgreSQL, GeoServer) in an Agile team to serve national scientific data through OGC-compliant APIs for the New Zealand Ocean Data Network (NZODN) portal.

Web and Content Intern - *Wellington Chamber of Commerce* 2021

- Developed dynamic web components and data visualisations for the iMIS engagement management system, automating the digitisation of paper-based member resources.

EDUCATION

Doctor of Philosophy, Artificial Intelligence - *Victoria University of Wellington* 2022 – 2026

- Thesis: *Machine Learning for Rapid Evaporative Ionization Mass Spectrometry for Marine Biomass Analysis*. Wellington Doctoral Scholarship and MBIE Doctoral Grant.
- Transformers reaching 100% accuracy on species identification and 74% on body-part classification against a 51% OPLS-DA baseline; ordinal models for graded oil contamination; transfer learning across tasks; *SpectroSim*, a self-supervised contrastive method that traces samples to their batch at 71% accuracy with no labels; decisions explained with LIME, SHAP and Grad-CAM.

Bachelor of Engineering (Honours), Software Engineering - *Victoria University of Wellington* 2016 – 2022

PUBLICATIONS

- Wood, J. (2026). Machine Learning for Rapid Evaporative Ionization Mass Spectrometry for Marine Biomass Analysis. PhD thesis, *Victoria University of Wellington*. doi:10.26686/mef0-mr46
- Wood, J., Nguyen, B., Xue, B., Zhang, M. and Killeen, D. (2025). Hook, line, and spectra: machine learning for fish species identification and body part classification using rapid evaporative ionization mass spectrometry. *Intelligent Marine Technology and Systems*, 3(1), 16. Springer. doi:10.1007/s44295-025-00066-3
- Wood, J., Nguyen, B. H., Xue, B., Zhang, M. and Killeen, D. (2022). Automated fish classification using unprocessed fatty acid chromatographic data: a machine learning approach. *Australasian Joint Conference on Artificial Intelligence (AJCAI)*, pp. 516–529. Springer. doi:10.1007/978-3-031-22695-3_36

OPEN SOURCE

- **ruru**, a plain-text literature library with feed triage, a terminal UI and an agent-friendly CLI; **taura**, a provenance ledger that checks every number, table and figure in a paper back to its run and commit; **gpu-spooler**, a job queue for SSH-reachable GPU servers used on the VUW ECS cluster.
- **qsh**, a local-first CLI that brings a small vision-language model to the Unix pipe (Rust, Python); **ZeroSlop**, a Chrome extension that flags AI-generated and engagement-farmed posts; **Cloudy with a Chance of Git Pulls**, a GitHub Action written up on HackerNoon.
- **Teaching Sand to Dream**, a book on how humans taught sand to dream, from Turing to ChatGPT, written without equations for curious non-experts.

TECHNICAL SKILLS

Machine learning: PyTorch, Transformers, Hugging Face, scikit-learn, NumPy, pandas, Optuna, DEAP, Weights & Biases
Evaluation and explainability: LIME, SHAP, Grad-CAM, conformal prediction, ordinal classification, rigorous statistical evaluation
Analytical data: REIMS and GC-MS mass spectrometry, chromatogram alignment, high-dimensional low-sample-size learning
Languages and systems: Python, Rust, C++, Haskell, TypeScript, SQL, Docker, Linux, Git, PostgreSQL, GeoServer, Bun

References available upon request.